

S66.

P1. Solution: $T=MN$

C is similar to A

P3. Solution:

$$\begin{aligned} \begin{bmatrix} 0 & 1 \\ 0 & 1 \end{bmatrix} &= \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \\ \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} &= \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \\ \begin{bmatrix} 4 & 3 \\ 2 & 1 \end{bmatrix} &= \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \end{aligned}$$

P5 Solution:

i) $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

ii) $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$

iii) $\begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 1 \end{bmatrix}$ similar.

P8. Solution: No.

$$A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, B = \begin{bmatrix} 0 & 2 \\ 0 & 0 \end{bmatrix}$$

P10. Solution:

$$J = \begin{bmatrix} c & 1 \\ 0 & c \end{bmatrix}$$

$$J^2 = \begin{bmatrix} c^2 & 2c \\ 0 & c^2 \end{bmatrix}$$

$$J^3 = \begin{bmatrix} c^3 & 3c \\ 0 & c^3 \end{bmatrix}$$

Guess $J^k = \begin{bmatrix} c^k & k c^{k-1} \\ 0 & c^k \end{bmatrix}$

$$J^0 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$J^{-1} = \begin{bmatrix} \frac{1}{c} & -\frac{1}{c^2} \\ 0 & \frac{1}{c} \end{bmatrix}$$

P17. (a) True

(b) False

(c) False

(d) True

P18. Proof: Suppose (λ, x) is an eigenvalue and corresponding eigenvector of AB .

Clearly $x \neq 0$. Then

$$ABx = \lambda x$$

$$\Rightarrow BABx = \lambda Bx$$

$$BAy = \lambda y$$

where $y = Bx \neq 0$ Since B is invertible.

$\Rightarrow (\lambda, Bx)$ is an eigenvalue and corresponding eigenvector of BA .

S7.1.

P1. Solution: $\frac{W = -V}{C = 0}$

P3. Solution:

(a), (b), (c) are linear.

P4. Solution:

(a) $S(T(v)) = V$

(b) $S(T(v_1 + v_2)) = S(T(v_1) + T(v_2))$
 $= S(T(v_1)) + S(T(v_2))$

P5. Solution:

(b), (c)

P7. Solution:

(a) $T(T(v)) = V$ linear

(b) $T(T(v)) = V + (2, 2)$, not linear

$$cc) T(T(v)) = T(-v_2, v_1) = (-v_1, -v_2)$$

linear

$$cd) T(T(v)) = T\left(\frac{v_1+v_2}{2}, \frac{v_1+v_2}{2}\right) \\ = \left(\frac{v_1+v_2}{2}, \frac{v_1+v_2}{2}\right)$$

linear.

P8. ca) Range of $T = \mathbb{R}^2$

kernel of $T = \{0\}$

cb) Range of $T = \mathbb{R}$

kernel of $T =$ The set of all vectors $(0, 0, v_3)$

cc) Range of $T = \{0\}$

kernel of $T = \mathbb{R}^2$

cd) Range of $T =$ The set of all vectors (v_1, v_1)

kernel of $T =$ The set of all vectors $(0, v_2)$

Pro. Solution:

ca) $T(v_1, v_2) = 0$
 $\Rightarrow (v_2, v_2) = 0$

$\Rightarrow v_2 = 0$

kernel of $T = \{(v_1, 0)\}$

cb) $T(v_1, v_2) = 0$

$\Rightarrow (v_1, v_2, v_1 + v_2) = 0$

$\Rightarrow v_1 = v_2 = 0$

kernel of $T = \{0\}$

cc) $T(v_1, v_2) = 0$

$\Rightarrow v_1 = 0$

$\Rightarrow$ kernel of $T = \{(0, v_2)\}$

P2. Solution:

$T(1, 1) = (2, 2), T(2, 0) = (0, 0)$

ca) $V = (2, 2)$

$T(V) = T(2, 2) = T(2(1, 1)) = 2T(1, 1) \\ = (4, 4)$

cb) $V = (3, 1)$

$T(3, 1) = T((1, 1) + (2, 0)) \\ = T(1, 1) + T(2, 0) \\ = (2, 2)$

cc) $V = (-1, 1)$

$T(-1, 1) = T((1, 1) - (2, 0)) \\ = T(1, 1) - T(2, 0) \\ = (2, 2)$

cd) $V = (a, b)$

$T(a, b) = T\left(\frac{b}{2}(1, 1) + \frac{a-b}{2}(2, 0)\right) \\ = \frac{b}{2}T(1, 1) + \frac{a-b}{2}T(2, 0) \\ = \frac{b}{2}(2, 2) + \frac{a-b}{2}(0, 0) \\ = (b, b)$

S7.2.

P1. Solution: $S(x) =$

$S(1) = \frac{d^2}{dx^2}(1) = 0, \frac{d^2}{dx^2}(x) = 0$

$S(x^2) = \frac{d^2}{dx^2}(x^2) = 2, \frac{d^2}{dx^2}(x^3) = 6x$

$\Rightarrow S(v_1) = 0w_1 + 0w_2 + 0w_3 + 0w_4$

$S(v_2) = 0w_1 + 0w_2 + 0w_3 + 0w_4$

$S(v_3) = 2w_1 + 0w_2 + 0w_3 + 0w_4$

$S(v_4) = 0w_1 + 6w_2 + 0w_3 + 0w_4$

$\Rightarrow B = \begin{bmatrix} 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 6 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

P3. Solution:

$A = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

$A^2 = \begin{bmatrix} 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 6 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

$= B$